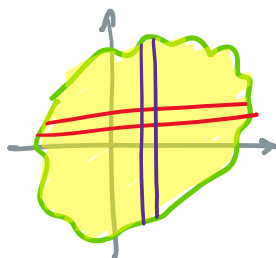


6.5: Continuous Conditional Distributions

For continuous random variables X and Y we can similarly define a conditional density function:

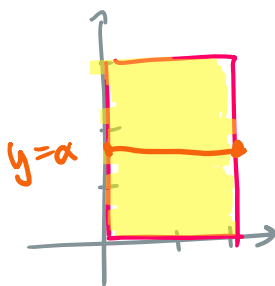
$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

This is shorthand for $P(x \leq X \leq x + dx | y \leq Y \leq y + dy)$



$$\begin{aligned} &P(x \leq X \leq x+dx | y \leq Y \leq y+dy) \\ &= \frac{P(x \leq X \leq x+dx, y \leq Y \leq y+dy)}{P(y \leq Y \leq y+dy)} \xrightarrow[\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}]{\frac{f_{XY}}{f_Y}} \end{aligned}$$

Example: Consider the joint density function $f(x, y) = \begin{cases} \frac{1}{18}xy^2 & 0 < x < 2, 0 < y < 3 \\ 0 & \text{else} \end{cases}$
for the random variables X, Y . Find $f_{X|Y}(x|\alpha)$ given $\alpha \in (0, 3)$

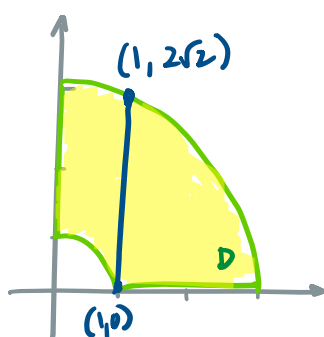


$$f_{X|Y}(x|\alpha) = \frac{f_{XY}(x, \alpha)}{f_Y(\alpha)} = \frac{\frac{1}{18}x\alpha^2}{\alpha^2/9} = \boxed{\frac{1}{2}x \quad 0 < x < 2}$$

$$\begin{aligned} f_Y(\alpha) &= \int_{-\infty}^{\infty} f_{XY}(x, \alpha) dx = \int_0^2 \frac{1}{18}x\alpha^2 dx \\ &= \frac{\alpha^2}{18} \int_0^2 x dx = \frac{\alpha^2}{18} \cdot \frac{x^2}{2} \Big|_0^2 = \frac{\alpha^2}{9} \end{aligned}$$

$$\int_0^2 \frac{1}{2}x dx = \frac{1}{4}x^2 \Big|_0^2 = 1 \quad \checkmark$$

Example: Consider the joint density function $f(x, y) = \begin{cases} \frac{c}{x^2+y^2} & 1 < x^2 + y^2 < 9, x, y \geq 0 \\ 0 & \text{else} \end{cases}$
for the random variables X, Y . Find $f_{Y|X}(y|1)$



$$\begin{aligned} 1 &= \iint_D f_{XY} dA = \int_0^{\pi/2} \int_1^3 \frac{c}{r^2} \cdot r dr d\theta \\ &= c \cdot \int_0^{\pi/2} d\theta \cdot \int_1^3 \frac{1}{r} dr = c \cdot \theta \Big|_0^{\pi/2} \cdot \ln|r| \Big|_1^3 \\ &= c \cdot \frac{\pi}{2} \cdot \ln|3| \quad \rightarrow c = \frac{2}{\pi \ln(3)} \end{aligned}$$

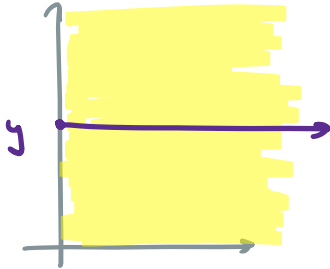
$$f_X(1) = \int_{-\infty}^{\infty} f_{XY}(1, y) dy = \int_0^{2\sqrt{2}} \frac{c}{1^2 + y^2} dy = c \cdot \tan^{-1}(y) \Big|_0^{2\sqrt{2}} = c \cdot \tan^{-1}(2\sqrt{2})$$

$$f_{Y|X}(y|1) = \frac{f_{XY}(1, y)}{f_X(1)} = \frac{\frac{c}{1^2 + y^2}}{c \cdot \tan^{-1}(2\sqrt{2})} = \boxed{\frac{1}{(1+y^2) \tan^{-1}(2\sqrt{2})} \quad 0 \leq y \leq 2\sqrt{2}}$$

Once again we can define conditional expectation as expected:

$$E[g(X)|Y=y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y) dx$$

Example: Let $f_{XY}(x,y) = \frac{1}{ye^{\frac{y^2+x}{y}}}$ for $x, y > 0$ and find $E[X|Y=y]$



$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{XY}(x,y) dx = \int_0^{\infty} \frac{1}{y} e^{-y-\frac{x}{y}} dx \\ &= \frac{1}{y} e^{-y} \int_0^{\infty} e^{-\frac{x}{y}} dx = \frac{1}{y} e^{-y} \left[\frac{e^{-x/y}}{-1/y} \right]_0^{\infty} \\ &= -e^{-y} [0 - 1] = e^{-y} \end{aligned}$$

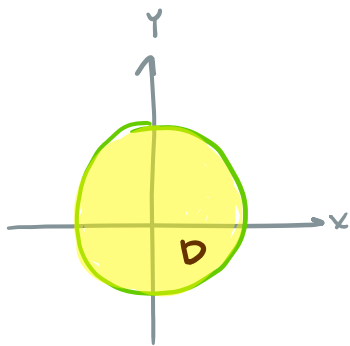
$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{\frac{1}{y} e^{-y-\frac{x}{y}}}{e^{-y}} = \frac{1}{y} e^{-x/y} \quad (x > 0)$$

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) dx = \int_0^{\infty} x \cdot \frac{1}{y} e^{-x/y} dx$$

$$\begin{aligned} &= \frac{1}{y} \int_0^{\infty} x e^{-x/y} dx = \frac{1}{y} \left(x \cdot \frac{e^{-x/y}}{-1/y} - \int \frac{e^{-x/y}}{y^2} dx \right) \Big|_0^{\infty} \\ &= \left[-x e^{-x/y} - y e^{-x/y} \right]_0^{\infty} \\ &= (-0 - 0) - (-0 - y \cdot 1) = y \end{aligned}$$

+	x	$e^{-x/y}$
-	1	$\frac{e^{-x/y}}{-1/y}$
	0	$\frac{e^{-x/y}}{y^2}$

Example: For joint density $f_{XY}(x, y) = ke^{-x^2-y^2}$ on $\mathbb{R}^2$, what is $E[X^2 | X^2 + Y^2 \leq 1]$?

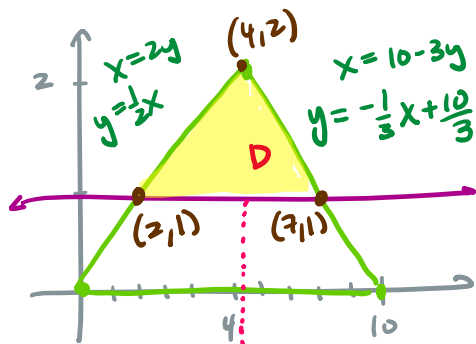


$$\begin{aligned}
 1 &= \iint_D f_{XY} dA = \int_0^{2\pi} \int_0^1 k \cdot e^{-r^2} r dr d\theta \\
 &= k \int_0^{2\pi} d\theta \cdot \int_0^1 e^{-r^2} r dr = k \cdot \theta \Big|_0^{2\pi} \cdot \left(-\frac{1}{2} e^{-r^2} \right) \Big|_0^1 \\
 &= k \cdot 2\pi \cdot \left(-\frac{1}{2} e^{-1} + \frac{1}{2} \right) = k \cdot \pi(1 - e^{-1}) \rightarrow k = \frac{1}{\pi(1 - e^{-1})}
 \end{aligned}$$

$$\begin{aligned}
 E[X^2 | X^2 + Y^2 \leq 1] &= \iint_D x^2 \cdot k e^{-x^2-y^2} dA = \int_0^{2\pi} \int_0^1 (r \cos \theta)^2 \cdot k \cdot e^{-r^2} r dr d\theta \\
 &= k \cdot \int_0^{2\pi} \cos^2 \theta d\theta \cdot \int_0^1 r^2 e^{-r^2} r dr \quad \left(\begin{array}{ll} w = -r^2 & w(0) = 0 \\ dw = -2r dr & w(1) = -1 \\ \frac{dw}{-2} = r dr & \end{array} \right) \\
 &= k \cdot \int_0^{2\pi} \frac{1}{2} (1 + \cos(2\theta)) d\theta \cdot \int_0^{-1} -w e^w \left(\frac{dw}{-2} \right) \\
 &= \frac{k}{4} \left(\theta + \frac{1}{2} \sin(2\theta) \right) \Big|_0^{2\pi} \cdot (w-1)e^w \Big|_0^{-1} \\
 &= \frac{k}{4} \cdot (2\pi + 0 - 0 - 0) \cdot [-2e^{-1} + 1e^0] = \frac{k\pi}{2} \cdot (1 - 2e^{-1}) \\
 &= \frac{1}{\pi(1 - e^{-1})} \cdot \frac{\pi}{2} (1 - 2e^{-1}) = \boxed{\frac{1 - 2e^{-1}}{2 - 2e^{-1}}}
 \end{aligned}$$

Example: Let $f_{XY}(x, y) = ky^2$ for (x, y) inside of the triangle with vertices $(0, 0)$, $(4, 2)$, $(10, 0)$ and zero elsewhere.

What is $E[X|Y \geq 1]$?



$$\begin{aligned}
 1 &= \iint_D ky^2 dA = \int_1^2 \int_{2y}^{10-3y} ky^2 dx dy \\
 &= k \int_1^2 y^2 x \Big|_{2y}^{10-3y} dy = k \int_1^2 y^2 (10-5y) dy \\
 &= k \int_1^2 (10y^2 - 5y^3) dy = k \cdot \left[\frac{10}{3} y^3 - \frac{5}{4} y^4 \right]_1^2 = k \cdot \frac{55}{12} \\
 &\quad \underline{\underline{k = \frac{12}{55}}}
 \end{aligned}$$

$$\begin{aligned}
 E[X|Y \geq 1] &= \iint_D x \cdot ky^2 dA = k \int_1^2 \int_{2y}^{10-3y} xy^2 dx dy \\
 &= k \int_1^2 \frac{1}{2} x^2 y^2 \Big|_{2y}^{10-3y} dy = \frac{k}{2} \int_1^2 [(10-3y)^2 - (2y)^2] \cdot y^2 dy \\
 &= \frac{k}{2} \int_1^2 (100y^2 - 6y^3 + 5y^4) dy = \frac{k}{2} \left[\frac{100}{3} y^3 - \frac{3}{2} y^4 + y^5 \right]_1^2 \\
 &= \frac{k}{2} \cdot \frac{118}{3} = \frac{6}{55} \cdot \frac{118}{3} = \boxed{\frac{236}{55}} \approx 4.29 \text{ something}
 \end{aligned}$$